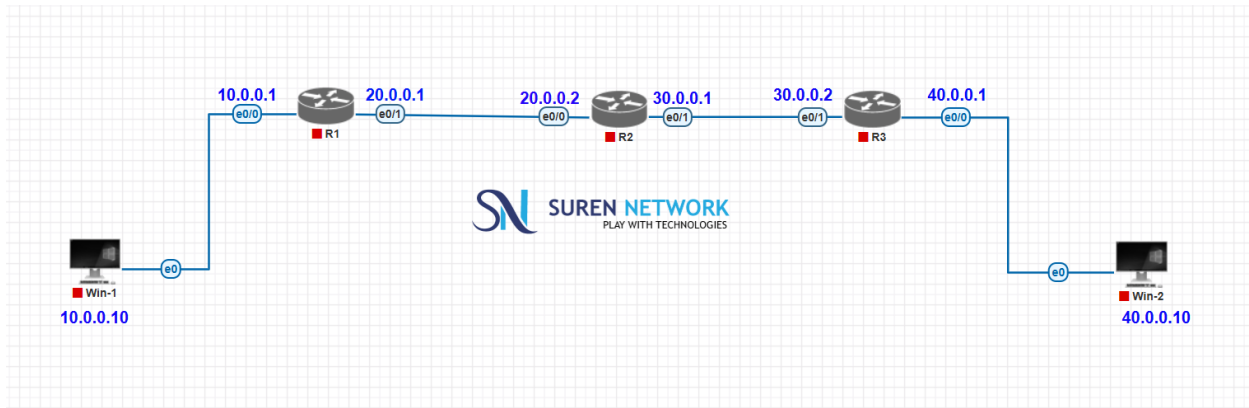


Static Route



Before the Ip configuration In PCs

```
QEMU (Win-1)
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . . : 
    Autoconfiguration IP Address. . . : 169.254.212.214
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 

C:\Documents and Settings\User>route print

Interface List
0x1 . . . . . MS TCP Loopback interface
0x2 . . . . . Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

Active Routes:
    Network Destination  Netmask          Gateway          Interface        Metric
    127.0.0.0             255.0.0.0        127.0.0.1       127.0.0.1        1
    169.254.0.0          255.255.0.0      169.254.212.214 169.254.212.214  10
    169.254.212.214      255.255.255.255  127.0.0.1       127.0.0.1        10
    169.254.255.255      255.255.255.255  169.254.212.214 169.254.212.214  10
    224.0.0.0            240.0.0.0        169.254.212.214 169.254.212.214  10
    255.255.255.255      255.255.255.255  169.254.212.214 169.254.212.214  1

Persistent Routes:
    None

C:\Documents and Settings\User>_

QEMU (Win-2)
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . . : 
    Autoconfiguration IP Address. . . : 169.254.211.141
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 

C:\Documents and Settings\User>route print

Interface List
0x1 . . . . . MS TCP Loopback interface
0x2 . . . . . Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

Active Routes:
    Network Destination  Netmask          Gateway          Interface        Metric
    127.0.0.0             255.0.0.0        127.0.0.1       127.0.0.1        1
    169.254.0.0          255.255.0.0      169.254.211.141 169.254.211.141  10
    169.254.211.141      255.255.255.255  127.0.0.1       127.0.0.1        10
    169.254.255.255      255.255.255.255  169.254.211.141 169.254.211.141  10
    224.0.0.0            240.0.0.0        169.254.211.141 169.254.211.141  10
    255.255.255.255      255.255.255.255  169.254.211.141 169.254.211.141  1

Persistent Routes:
    None

C:\Documents and Settings\User>
```

After the Ip configuration In PCs

```
QEMU (Win-1)
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . . : 
    IP Address . . . . . : 10.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.1

C:\Documents and Settings\User>route print

Interface List
0x1 . . . . . MS TCP Loopback interface
0x2 . . . . . Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

Active Routes:
    Network Destination  Netmask          Gateway          Interface        Metric
    0.0.0.0              0.0.0.0          10.0.0.1       10.0.0.10        10
    10.0.0.0             255.0.0.0        10.0.0.10      10.0.0.10        10
    10.0.0.10           255.255.255.255  127.0.0.1       127.0.0.1        10
    10.255.255.255       255.255.255.255  10.0.0.10      10.0.0.10        10
    127.0.0.0            255.0.0.0        127.0.0.1       127.0.0.1        1
    224.0.0.0            240.0.0.0        10.0.0.10      10.0.0.10        10
    255.255.255.255      255.255.255.255  10.0.0.10      10.0.0.10        1
    Default Gateway:    10.0.0.1

Persistent Routes:
    None

C:\Documents and Settings\User>_

QEMU (Win-2)
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . . : 
    IP Address . . . . . : 40.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 40.0.0.1

C:\Documents and Settings\User>route print

Interface List
0x1 . . . . . MS TCP Loopback interface
0x2 . . . . . Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

Active Routes:
    Network Destination  Netmask          Gateway          Interface        Metric
    0.0.0.0              0.0.0.0          40.0.0.1       40.0.0.10        10
    40.0.0.0             255.0.0.0        40.0.0.10      40.0.0.10        10
    40.0.0.10           255.255.255.255  127.0.0.1       127.0.0.1        10
    40.255.255.255       255.255.255.255  40.0.0.10      40.0.0.10        10
    127.0.0.0            255.0.0.0        127.0.0.1       127.0.0.1        1
    224.0.0.0            240.0.0.0        40.0.0.10      40.0.0.10        10
    255.255.255.255      255.255.255.255  40.0.0.10      40.0.0.10        1
    Default Gateway:    40.0.0.1

Persistent Routes:
    None

C:\Documents and Settings\User>
```

Assign ip address in all cisco routers

```
R1
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int e0/0
R1(config-if)#ip add 10.0.0.1 255.0.0.0
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#
R1(config)#int e0/1
R1(config-if)#ip add 20.0.0.1 255.0.0.0
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#
R1(config)#do sh ip add br
      ^
% Invalid input detected at '^' marker.

R1(config)#do sh ip int br
Interface                IP-Address      OK? Method Status    Prot
ocol
Ethernet0/0              10.0.0.1        YES manual  up        up
Ethernet0/1              20.0.0.1        YES manual  up        up
Ethernet0/2              unassigned      YES TFTP    down      down
Ethernet0/3              unassigned      YES TFTP    down      down
R1(config)#
```

```
R2
R2(config)#
R2(config)#int e0/0
R2(config-if)#ip add 20.0.0.2 255.0.0.0
R2(config-if)#no sh
R2(config-if)#exi
R2(config)#
R2(config)#int e0/1
R2(config-if)#ip add 30.0.0.1 255.0.0.0
R2(config-if)#no sh
R2(config-if)#exit
R2(config)#
R2(config)#do sh ip int br
Interface                IP-Address      OK? Method Status    Prot
ocol
Ethernet0/0              20.0.0.2        YES manual  up        up
Ethernet0/1              30.0.0.1        YES manual  up        up
Ethernet0/2              unassigned      YES TFTP    down      down
Ethernet0/3              unassigned      YES TFTP    down      down
R2(config)#
```

```
R3
R3(config)#int e0/0
R3(config-if)#ip add 40.0.0.1 255.0.0.0
R3(config-if)#no sh
R3(config-if)#ex
R3(config)#
R3(config)#int e0/1
R3(config-if)#ip add 30.0.0.2 255.0.0.0
R3(config-if)#no sh
R3(config-if)#exi
R3(config)#
R3(config)#do sh ip int br
Interface                IP-Address      OK? Method Status    Prot
ocol
Ethernet0/0              40.0.0.1        YES manual up        up
Ethernet0/1              30.0.0.2        YES manual up        up
Ethernet0/2              unassigned      YES TFTP  down      down
Ethernet0/3              unassigned      YES TFTP  down      down
```

```
R1
Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
R1(config)#
```

```
R2
Gateway of last resort is not set

  20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/0
L       20.0.0.2/32 is directly connected, Ethernet0/0
  30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.0.0.0/8 is directly connected, Ethernet0/1
L       30.0.0.1/32 is directly connected, Ethernet0/1
R2(config)#
```

```
R3
Gateway of last resort is not set

  30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.0.0.0/8 is directly connected, Ethernet0/1
L       30.0.0.2/32 is directly connected, Ethernet0/1
  40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       40.0.0.0/8 is directly connected, Ethernet0/0
L       40.0.0.1/32 is directly connected, Ethernet0/0
R3(config)#
```

Add static route for missing route in router-1

```
R1
R1(config)#ip route 30.0.0.0 255.0.0.0 20.0.0.2
R1(config)#ip route 40.0.0.0 255.0.0.0 20.0.0.2
R1(config)#
R1(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
```

Add static route for missing route in router-2

```
R2
Router(config)#ip ro
Router(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
Router(config)#ip route 40.0.0.0 255.0.0.0 30.0.0.2
Router(config)#
```

Add static route for missing route in router-3

```
R3
R3(config)#ip route 20.0.0.0 255.0.0.0 30.0.0.1
R3(config)#ip route 10.0.0.0 255.0.0.0 30.0.0.1
R3(config)#
R3(config)#
```

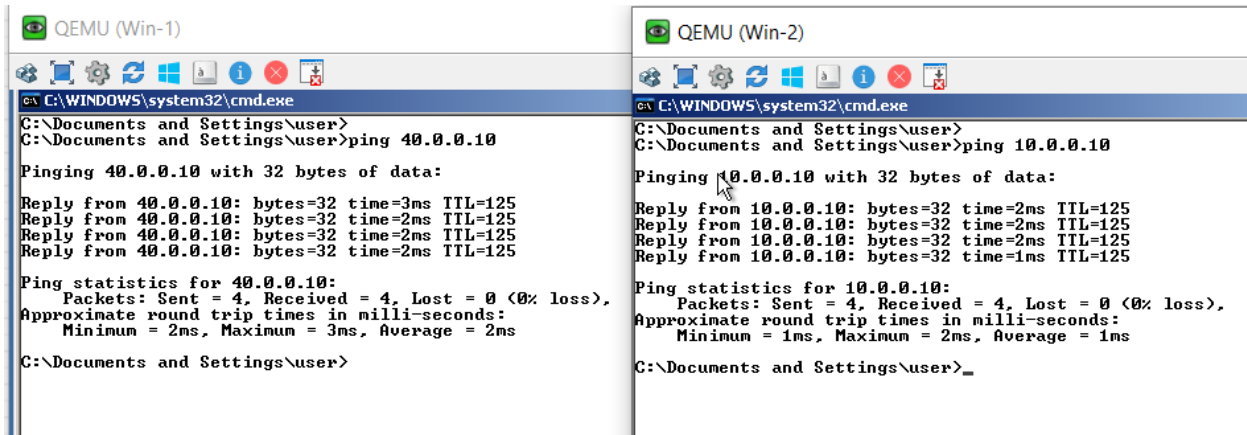
Check routing table for new routes in routing table

```
R1
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/8 is directly connected, Ethernet0/0
L    10.0.0.1/32 is directly connected, Ethernet0/0
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/1
L    20.0.0.1/32 is directly connected, Ethernet0/1
S    30.0.0.0/8 [1/0] via 20.0.0.2
S    40.0.0.0/8 [1/0] via 20.0.0.2

R2
S    10.0.0.0/8 [1/0] via 20.0.0.1
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/0
L    20.0.0.2/32 is directly connected, Ethernet0/0
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.1/32 is directly connected, Ethernet0/1
S    40.0.0.0/8 [1/0] via 30.0.0.2

R3
S    10.0.0.0/8 [1/0] via 30.0.0.1
S    20.0.0.0/8 [1/0] via 30.0.0.1
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.2/32 is directly connected, Ethernet0/1
40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    40.0.0.0/8 is directly connected, Ethernet0/0
L    40.0.0.1/32 is directly connected, Ethernet0/0
```

Ping 10.0.0.10 to 40.0.0.10



The image displays two side-by-side screenshots of QEMU virtual machines, labeled 'QEMU (Win-1)' and 'QEMU (Win-2)'. Both windows show a Windows command prompt running a ping command. The left window (Win-1) shows a ping from 40.0.0.10 to 40.0.0.10, while the right window (Win-2) shows a ping from 10.0.0.10 to 10.0.0.10. Both pings are successful with 0% loss and show round trip times in milliseconds.

```
QEMU (Win-1)
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>
C:\Documents and Settings\user>ping 40.0.0.10
Pinging 40.0.0.10 with 32 bytes of data:
Reply from 40.0.0.10: bytes=32 time=3ms TTL=125
Reply from 40.0.0.10: bytes=32 time=2ms TTL=125
Reply from 40.0.0.10: bytes=32 time=2ms TTL=125
Reply from 40.0.0.10: bytes=32 time=2ms TTL=125
Ping statistics for 40.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms
C:\Documents and Settings\user>
```

```
QEMU (Win-2)
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>
C:\Documents and Settings\user>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
Reply from 10.0.0.10: bytes=32 time=1ms TTL=125
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Documents and Settings\user>
```